



THE UNIVERSITY *of* EDINBURGH
informatics

Semantic Understanding in Legal NLP

Claire Barale

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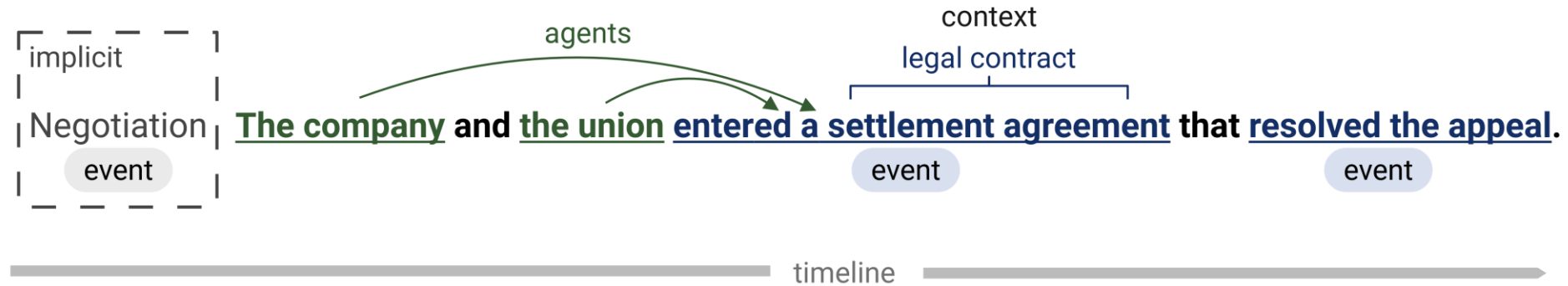
Why legal NLP?

Document
density

Reasoning
framework

Cross-domain
generalization

Understanding specialized language



Center-embedded clause

A transfer into settlement, whether revocable or irrevocable, is a disposal of the entire property thereby becoming settled property notwithstanding that the transferor has some interest as a beneficiary under the settlement and notwithstanding that he is a trustee, or the sole trustee, of the settlement.

Center-embedded clause

Domain alignment in pretraining

LexGLUE (Chalkidis et al., 2022)

LegalBench (Guha et al., 2023)

LegalAgentBench (Li et al., 2025)

→ LLMs performance is suboptimal in legal NLP tasks

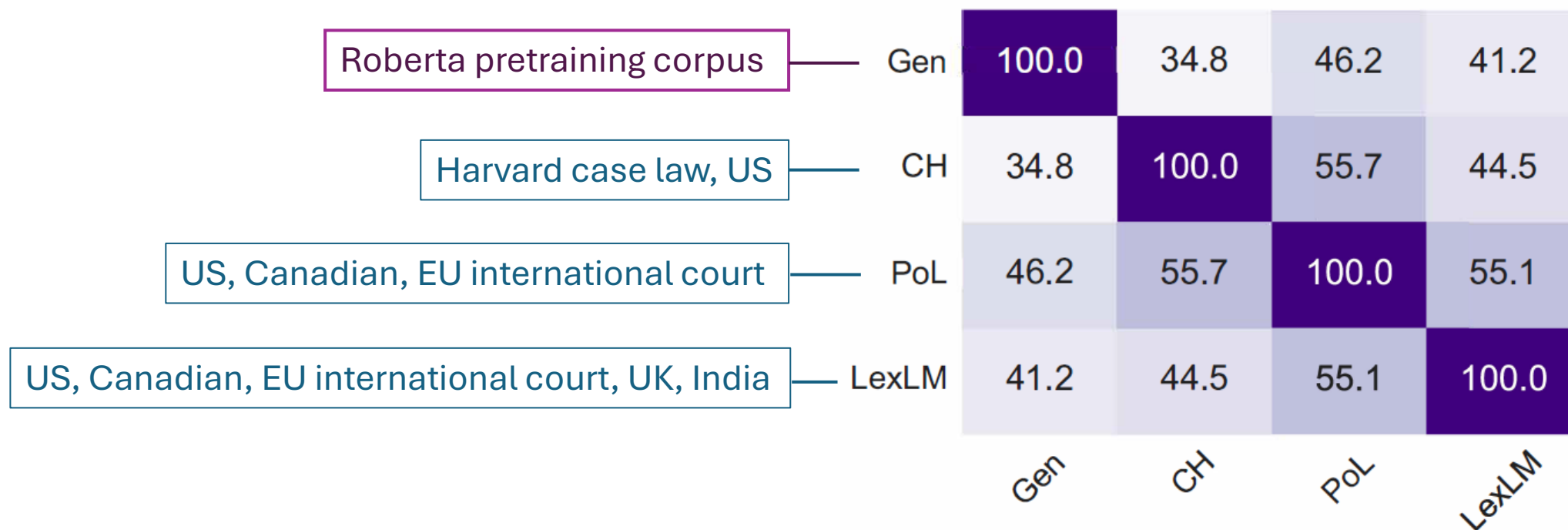


Figure: Vocabulary overlap (%) between the pretraining corpora. Gen stands for Generic and is sampled from sources similar to RoBERTa's pretraining corpus. Vocabularies are created with the top 10K most frequent tokens in a sample of 50K documents per model.

Hypothesis and outline of the talk

[Hypothesis] Advancements in legal NLP/NLU benefit from **explicit semantic structure** (entities, events, temporal relations) and explicit modeling of the linguistic specificities of legal discourse.

1. Where LLMs fail in legal NLP – Probing task

- Zero-shot LLMs struggle with legal entity understanding
- Preference for surface-level semantics over syntax

2. Open challenges in complex reasoning – Event ordering task

- Entity-level temporal reasoning remains challenging
- Legal language structure is a core failure mode

Behavioral probing

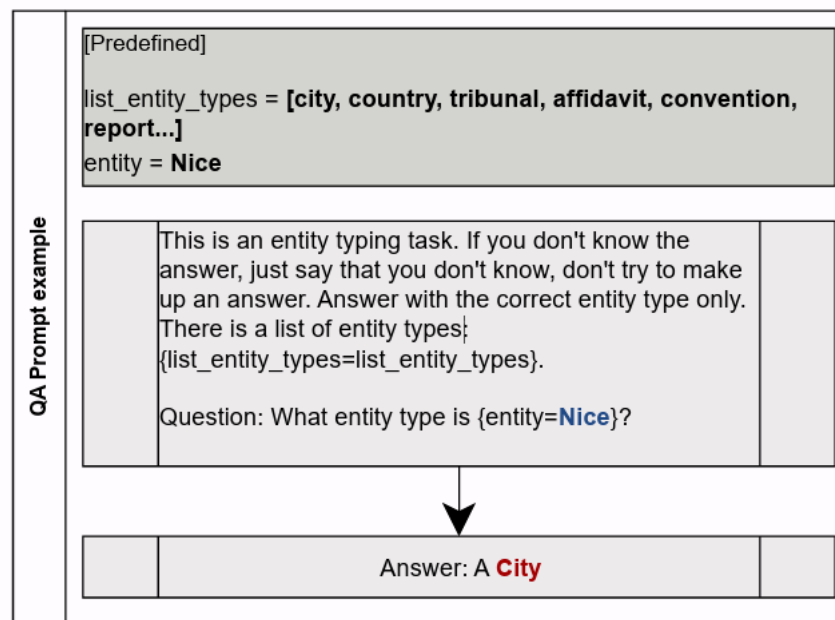
Probing latent legal knowledge in language models

[RQ] How proficient are language models at acquiring knowledge about domain-specific entities like legal entities during pretraining?

- Language models learn the probabilistic structure of a language
- One hypothesis is that LMs acquire diverse linguistic and semantic knowledge during pretraining (Petroni et al, 2019, Jiang et al., 2020)
- Domain-oriented pretraining and task-specific pretraining can enhance a model's performance and adaptability (Gururangan et al., 2020)

Probing legal entity types

- Task: Entity typing (predict the entity type), in a zero-shot setting, as a Multiple-Choice QA task
- Model: Llama-2-chat
- Dataset: Asylex, a dataset of refugee status determination decisions (Barale et al., 2023)



(b) *Experiment Llama2* QA prompt example

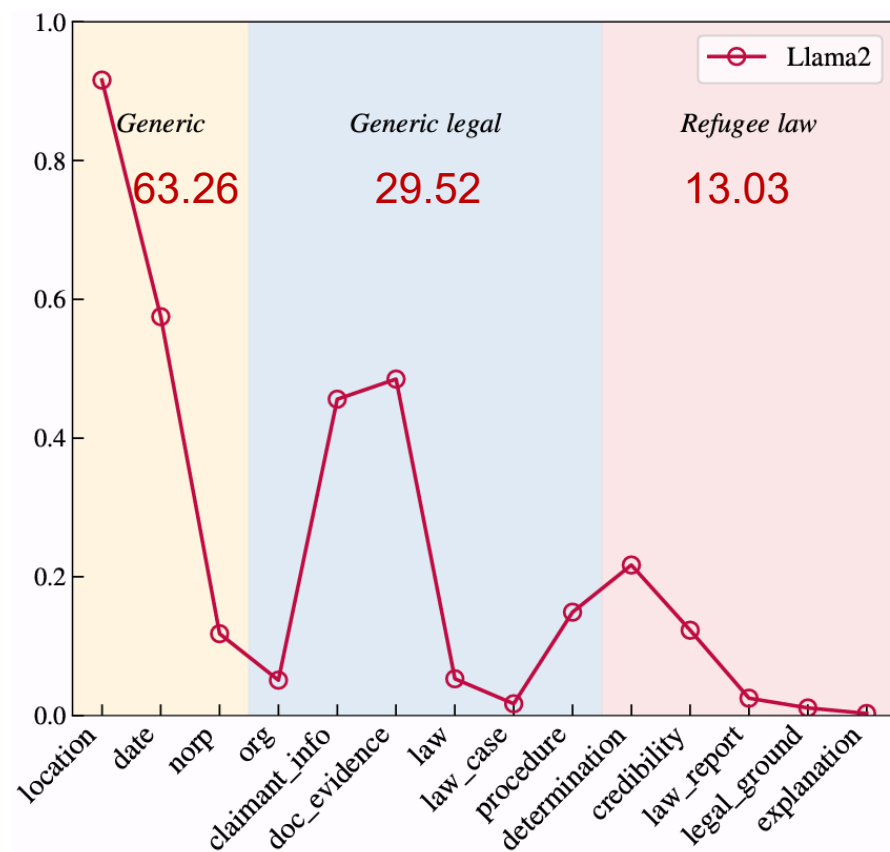
Fine-grained entity types – AsyLex dataset

Type	Description	Examples
LOCATION	Cities, countries, regions	Toronto, Ontario
DATE	Absolute or relative dates or periods	June 4, 1996; two years
NORP	Nationalities, religious, political, or ethnic groups	Hutu; Nigerian; Christian
ORG	Tribunals, companies, NGOs	Immigration Appeal Division
CREDIBILITY	Mentions of credibility	Lack of evidence; inconsistencies
DETERMINATION	Outcome of the decision (accept/reject)	Appeal is dismissed; not a Convention refugee
CLAIMANT_INFO	Claimant's attributes (age, gender, citizenship, occupation)	28-year-old; citizen of Iran; female
PROCEDURE	Steps in the claim and legal procedure	Removal order; sponsorship application
DOC_EVIDENCE	Evidence, proofs, supporting documents	Passport; medical record; marriage certificate
EXPLANATION	Reasons given by the panel for the determination	Fear of persecution; no protection by the state
LEGAL_GROUND	Convention grounds for refugee status	Political opinion; religion
LAW	Citations: legislation and international conventions	Section 1(a) of the Convention
LAW_CASE	Citations: case law and past decisions	X v. Minister of Canada, 1994
LAW_REPORT	Country reports (NGOs, UN, etc.)	Police and military torture of women in Mexico

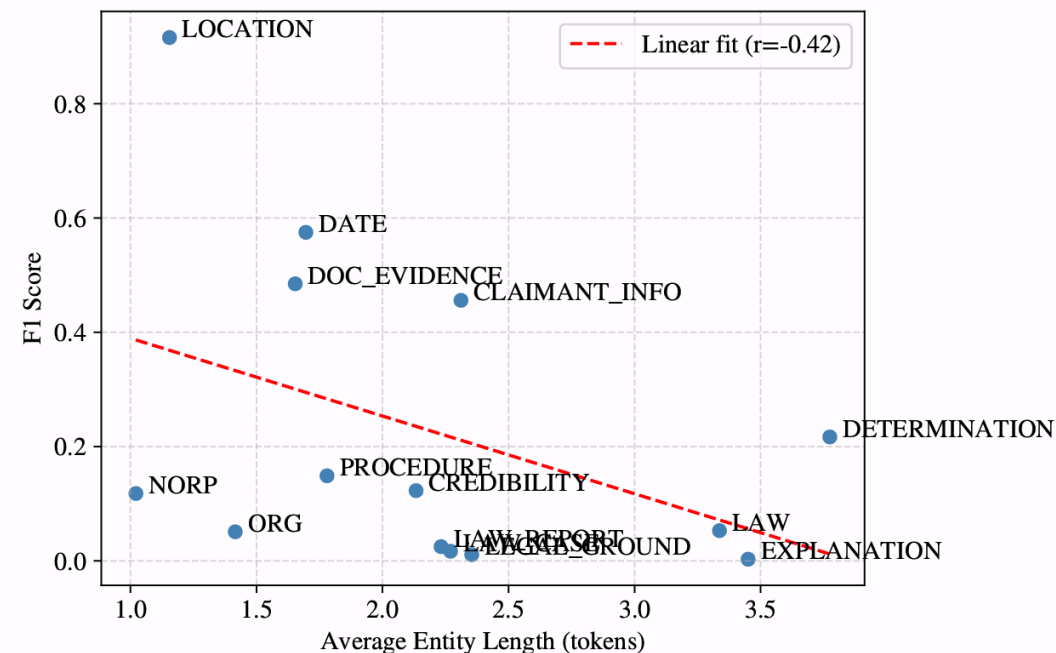
Table 3.2: Predefined fine-grained legal entity types used in our annotation schema.

Claire Barale, Mark Klaisoongnoen, Pasquale Minervini, Michael Rovatsos, and Nehal Bhuta. *AsyLex: A Dataset for Legal Language Processing of Refugee Claims*. NLLP 2023, ACL.

Experimental results



F1 scores for entity type prediction across models, with entity types ordered by increasing domain complexity.



(a) All entities included.

Correlation between entity average length and F1 score for the Llama2 model.

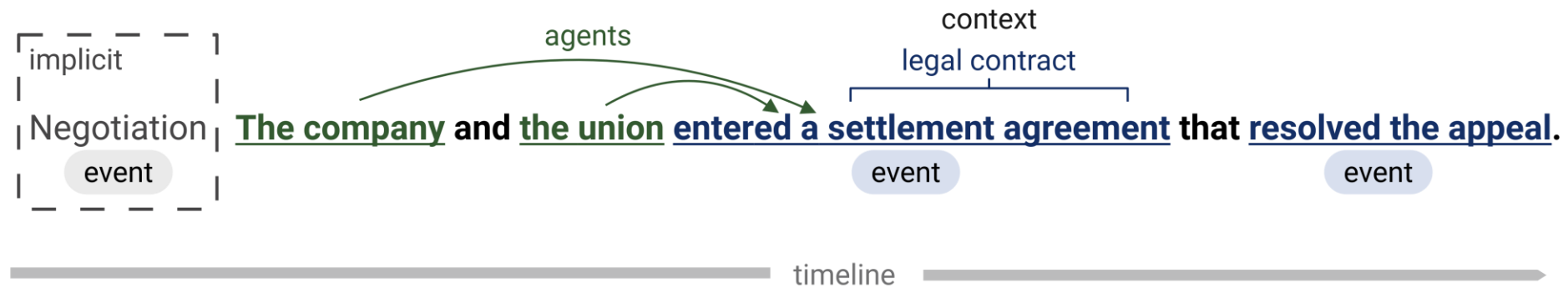
Experimental results – Error analysis

	Error Type	Prompt example	Prediction	Gold	%
Llama2	Arbitrary prediction	What is <i>Subsection 648</i> ?	country	LAW	22.22
	Closely related	What is <i>vietnamese</i> ?	country	nationality (NORP)	18.52
	False negative	What is <i>female claimant</i> ?	female claimant	gender (CLAIMANT_INFO)	33.33
	Prompt error	What is <i>removal order</i> ?	It is a type of judicial decision.	PROCEDURE	25.93

From entities to events

Legal reasoning operates over events, and their relations -- not just static entities

Events embed entities + actions + temporal relations

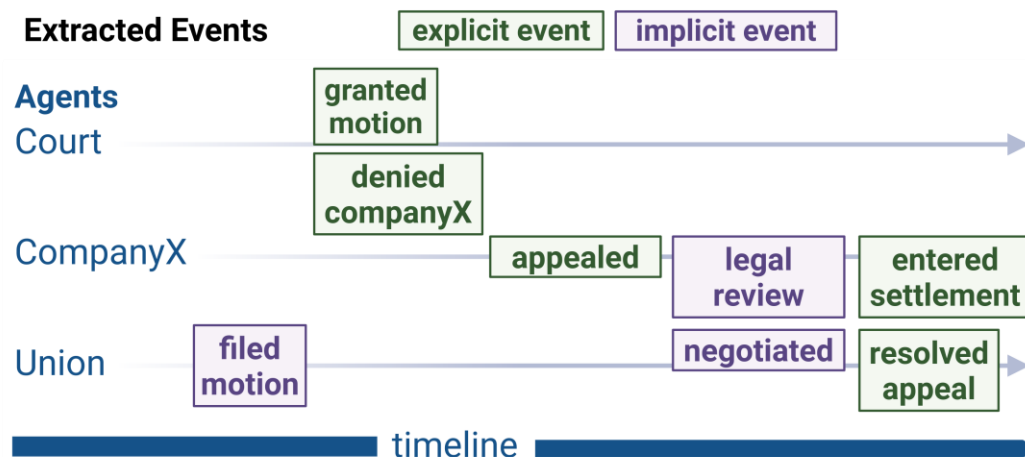


Task overview: Event ordering on the LexTime dataset

Context paragraph

On November 25, 2020, the district court **granted** the unions summary judgment motion and **denied** CompanyX. CompanyX subsequently **appealed** the matter to the United States Court of Appeals for the Sixth Circuit (no. 20-xxxx). on September 17, 2021, however, the company and the union **entered** a settlement agreement that **resolved** the appeal.

Extracted Events



LexTime instance

Does the event **entering a settlement agreement** **follow** companyX **appeal**? temporal relationship **Label**
Yes

Event-centric reasoning:

- states/actions situated in time
- temporal relations (before / after / simultaneous)

Labor-related federal complaints (2020-2024)

512 manually annotated instances, balanced (yes/no labels)

Two event types:

- Explicit events: stated directly (e.g., *granted motion*)
- Implicit events: inferred from context (e.g., *negotiated*)

Query format:

{event A, temporal relation, event B}
→ “Does A happen before B?”

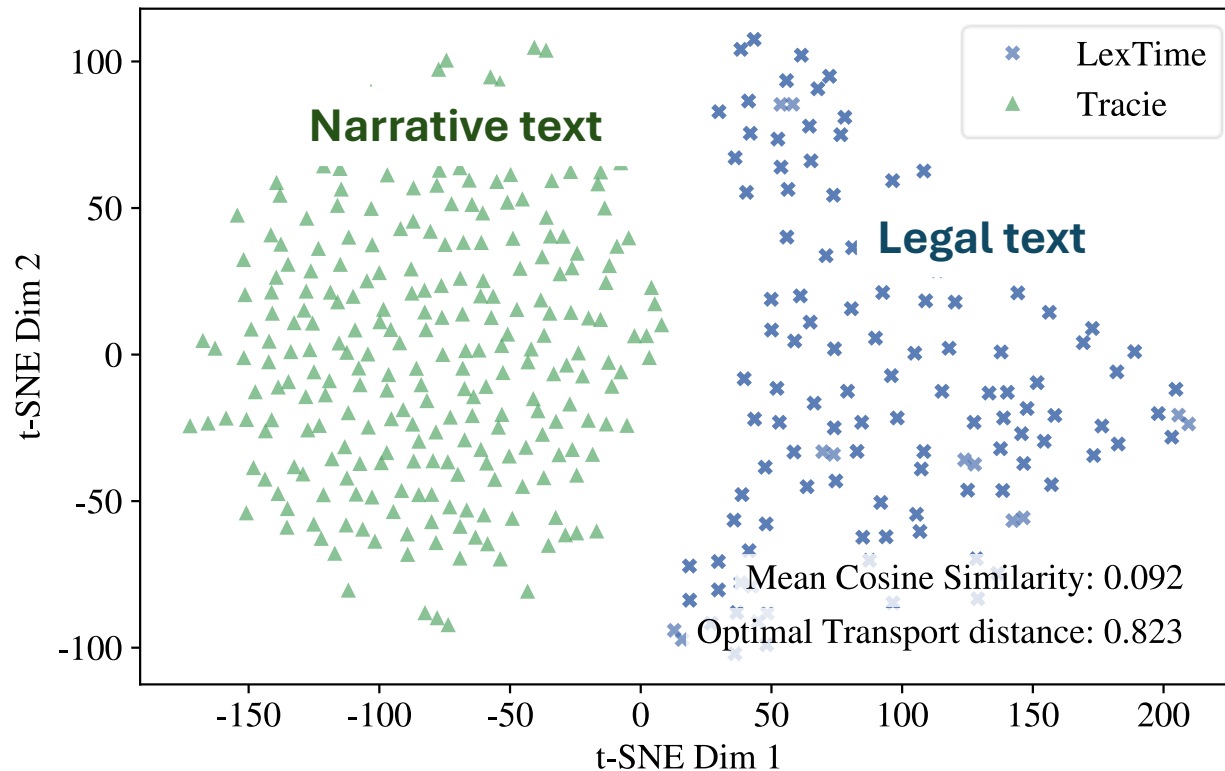
Comparative Dataset: TRACIE, narrative benchmark

Temporal Reasoning on Implicit Events from Distant Supervision

Ben Zhou^{1,2} Kyle Richardson¹ Qiang Ning³ Tushar Khot¹ Ashish Sabharwal¹ Dan Roth²

^{*}¹Allen Institute for AI ²University of Pennsylvania ³Amazon

Specialized language: Legal language is a linguistic genre and register



Visualization of the split of paragraph embeddings containing events using t-SNE (Van der Maaten and Hinton, 2008).

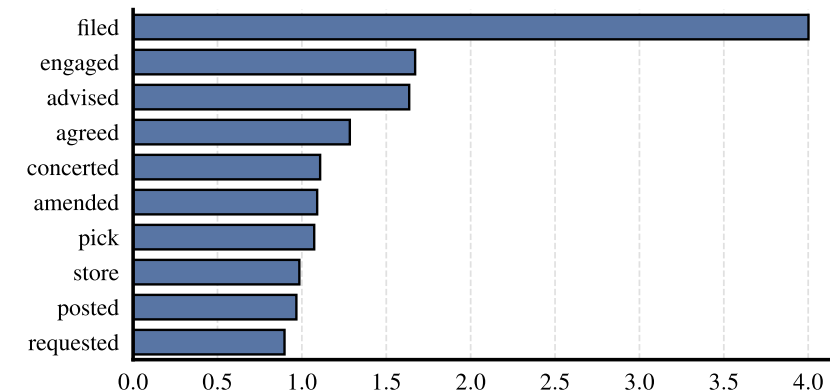
Low mean cosine similarity indicates that paragraphs are semantically different on average.

(RQ) How do language models react to the unique features of legal language for event ordering?

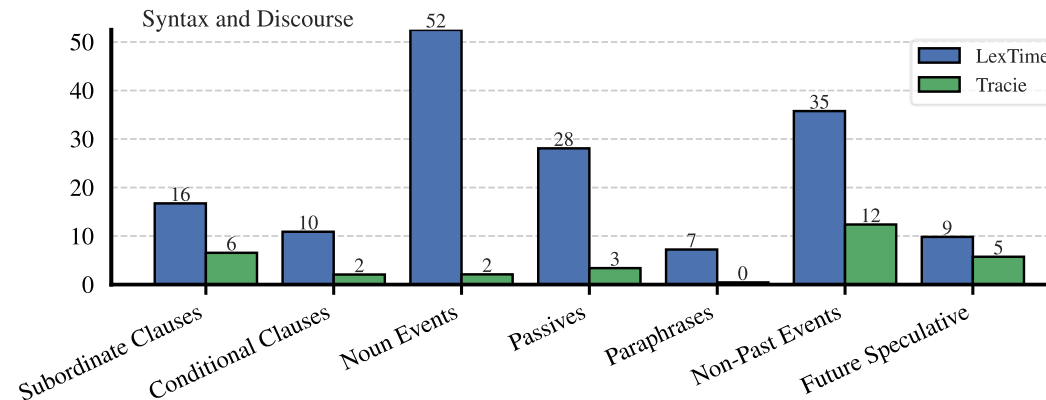
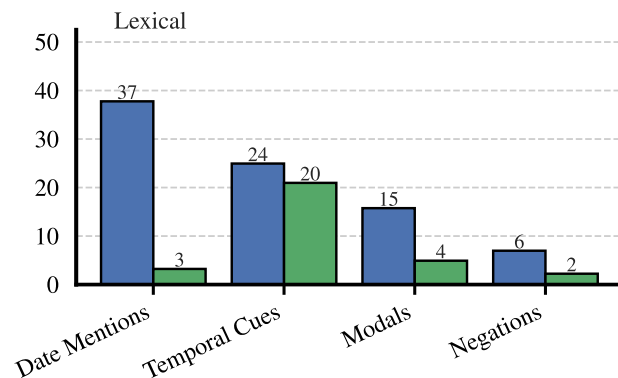
Legal language: Distinct distribution

Lexical and syntactic differences (LexTime vs. Tracie)

- Longer sentences (28 vs 9 tokens, median)
- Specialized legal vocabulary (*filed*, *granted*, *amended*)
- Higher event density (4.01 vs 1.60 events/sentence)
- More subordination (+10.5%) and modality (+11%)
- Frequent nominalization (+50.4%), paraphrasing (+6.56%), passive voice (+24.76%)

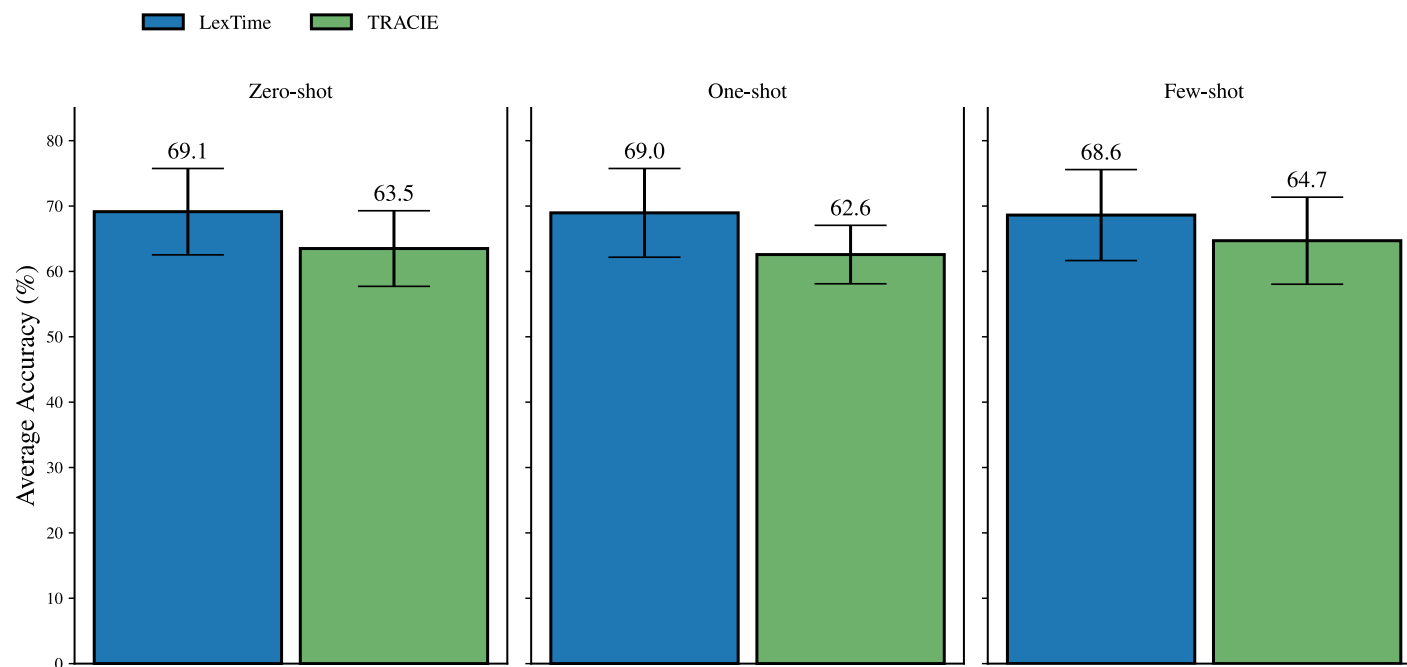


Top-10 event triggers in LexTime by frequency (%).



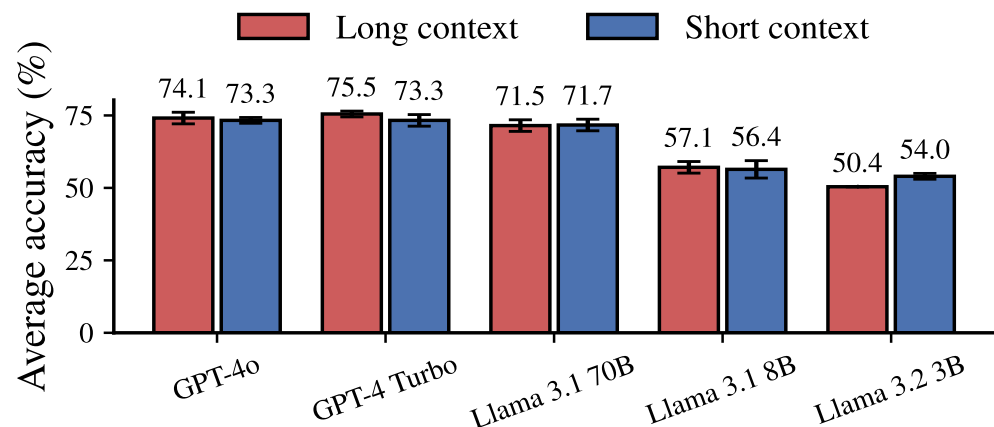
Comparison of linguistic features in LexTime and Tracie. Values represent the percentage of sentences that contain each features, indicating frequency in the text.

Result 1: Models achieve higher event ordering accuracy on LexTime than on TRACIE, by up to 10.5% (GPT-4o, few-shot)



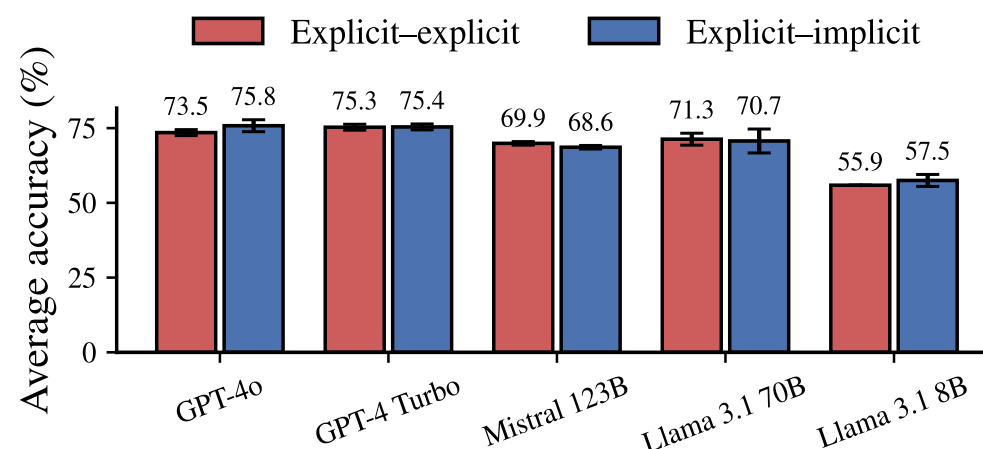
Results are averaged from GPT-4, GPT4-Turbo, Llama 3.1_{70B}, Llama 3.1_{8B} and Mistral_{123B}. 3 runs.

Result 2: Large models generally benefit from longer context windows



Context length (short <150 tokens, long ≥150).

Result 3: Large models perform better when implicit events are included



Explicit vs. implicit event pairs.

Query: Does event A happen before event B?

Example explicit-explicit pair: {explicit event A, TR, explicit event B}

Example explicit-implicit pair: {explicit event A, TR, implicit event B}

Results are averaged from zero-shot, one-shot, few-shot settings. 3 runs.

Result 4: Chain-of-thought prompting is not effective

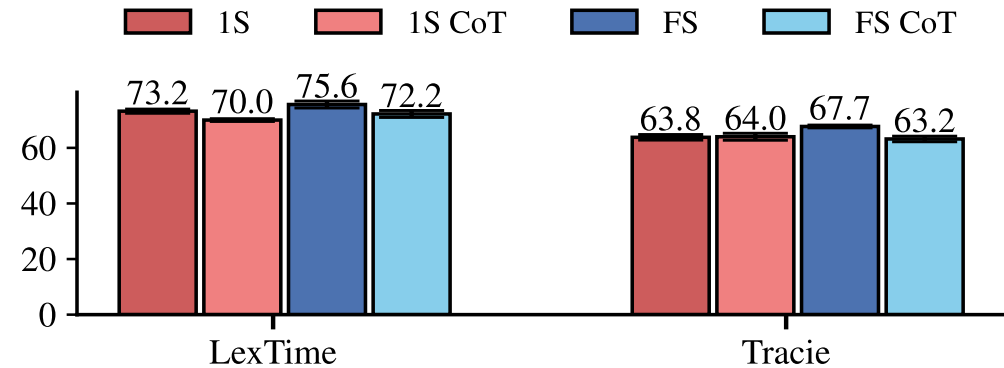
Example used for CoT prompts

{Context} *Before assigning the plaintiff to work, the defendant hired employees, including the plaintiff. The plaintiff was hired as a trackman.*

{Query} *Does hiring employees happens before assigning the plaintiff to work?*

{Answer} *Reasoning: 1. The defendant hired employees, including the plaintiff.
2. The plaintiff was hired as a trackman.
3. The employees including the plaintiff were all assigned to work.*

The correct answer is: "yes"



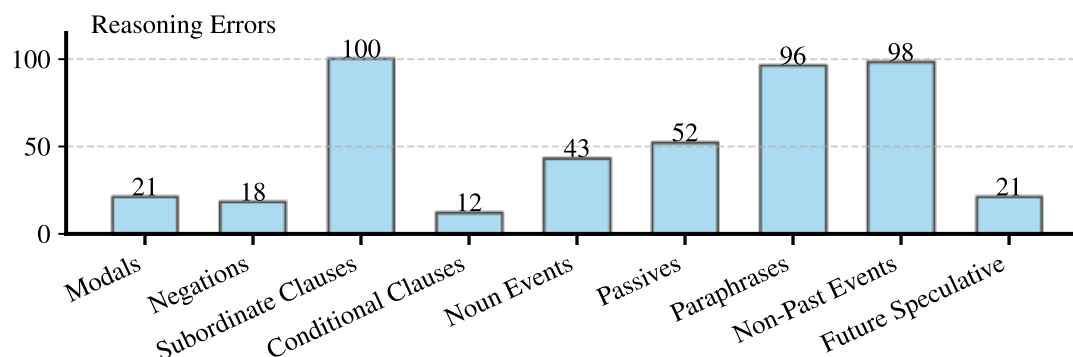
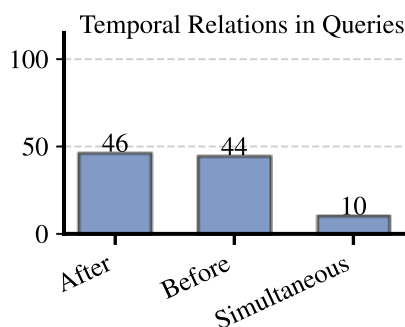
Performance gap (accuracy) with and without CoT prompting.

Results are averaged from GPT-4, GPT4-Turbo, Llama 3.1_{70B} and Mistral_{123B}. 3 runs.

Reasoning errors are associated with the linguistic specificities of legal language.

Compared to the overall dataset, errors are associated with:

- +89% increase in paraphrases
- +84% in subordinate clauses
- +63% in events not in the past simple tense
- +24% in passive voice
- +12% in negations and future speculative constructions
- +6% in modality markers
- +2% in conditional clauses.



Error analysis for LexTime on 100 misclassified samples from the few-shot setting across GPT-4, GPT-4 Turbo and Llama 3.1_{70B}.

Takeaways

Discussion & Implications

- LexTime introduces temporal reasoning to legal NLP.
- LLMs show improved performance on legal event ordering in LexTime compared to narrative, but linguistic complexities commonly found in legal language contribute to model errors.
- Calls for domain-sensitive modeling, tailored to specialized language types, and possibly data simplification.
- Opens directions for domain adaptation and cross-domain generalization.

Limitations & Future Work

- Only 512 samples and limited temporal relations schema (before/after/simultaneous).
- Focus on verb event triggers (noun event triggers are not annotated).
- Future work: extend to richer temporal structures, and across specialized domains.

